Stability Analysis for Feedforward Approximations in the Control of Flexible Joint Robots

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Abstract

In this paper we present a stability analysis w.r.t. different forms of feedforward. This aspect is of interest in the tracking problem for flexible joint robots when the controller is the sum of a model–based feedforward and a model–independent feedback. Choosing a particular feedforward as reference, for which local exponential stability is proven, the other proposed feedforwards can be seen as perturbations and stability properties can be inferred.

1 Introduction

A new approach in the control of flexible joint robots which exploits the inherent passivity property of the mechanical structure has recently been presented for both the set–point and the tracking problems [5, 7]. The proposed controller is the sum of a model–based feedforward and a model–independent feedback. The feedforward design is similar to the exact linearization approach, both essentially solve an inverse plant problem, but can requires much less model information, and the additive separation between the feedback and feedforward implies that errors in the feedforward do not lead to instability. Applying the feedforward leads to a system defined as error system which has to be stabilized. It is possible to prove that with the addition of a motor position error feedback the resulting system is passive and zero–state detectable for an appropriate choice of the output [5, 7]. It is also known [1] that a zero–state detectable and passive system can be asymptotically stabilized (eventually globally) by means of an appropriate output feedback. This procedure has been applied to flexible joint robots for both the set–point and tracking problem obtaining, for a large class of flexible joint robots, global asymptotic stability even in the absence of damping. However this approach does not give any information about local exponential stability, which would allow to draw conclusions on robustness.

Different possible forms for the feedforward can be devised depending on the available measurements and knowledge of the system's parameters. In this paper we will analyse the different forms of stability obtainable for each choice of the feedforward, leaving to [7] the study of global asymptotic stabilization for some particular choices of the feedforward. A robustness analysis with respect to variations in the feedforward is performed, extending [10] to the flexible joint case. In particular two specific forms are chosen as canonical feedforwards while the others are seen as variations. After a description in Section 2 of the dynamical model, we present in Section 3 a stability analysis under feedforward approximation. At first local exponential stability is proven for the canonical feedforwards, from which a stability analysis for the other forms of feedforward is performed.

2 Flexible joint manipulators modeling and general properties

Consider a flexible joint robot with the rotors of the actuators modeled as uniform bodies having their center of mass on the rotation axis. The dynamical model is given by [3, 8, 9]

$$M(\theta)\ddot{\theta} + C(\theta,\dot{\theta})\dot{\theta} + D(\dot{\theta}) + r(\theta) = Bu \qquad (2.1)$$

with $\theta^T = [\theta_\ell{}^T, \theta_m{}^T] \in \mathbf{R}^{2\cdot m}$, and

$$M(\theta) = \begin{bmatrix} M_{11}(\theta_\ell) & M_{12}(\theta_\ell) \\ M_{12}^T(\theta_\ell) & M_{22} \end{bmatrix}, \quad D(\dot{\theta}) = \begin{bmatrix} D_\ell(\dot{\theta}) \\ D_m(\dot{\theta}) \end{bmatrix}$$

$$B = \begin{bmatrix} 0_{m\times m} \\ I_{m\times m} \end{bmatrix} \quad r(\theta) = \begin{bmatrix} g_\ell(\theta_\ell) \\ 0 \end{bmatrix} + \begin{bmatrix} Nk_1(\theta_\ell,\theta_m) \\ -k_1(\theta_\ell,\theta_m) \end{bmatrix} \qquad (2.2)$$

θ_ℓ, θ_m are the link and motor displacement; $u \in \mathbf{R}^m$ is the input torque; M_{11} is the mass–inertia matrix of the links including the mass of the motors mounted

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on the links as if they were rigidly attached; M_{22} is the motor inertia matrix; M_{12} represents the coupling between the motor and link accelerations; D_ℓ and D_m are respectively the torques due to link and motor viscous and Coulomb friction (the effect of friction in the transmission element is neglected); $C(\theta,\dot\theta)$ represents the torques due to centrifugal and Coriolis forces; g_ℓ are the gravity forces acting upon the link; k_1 are the spring coupling (eventually nonlinear) forces while N is the gear ratios matrix.

If the stator of motor i is mounted on link $i-1$ while its rotor is elastically attached to link i for $i=1,\cdots,m$, M_{12} is strictly upper triangular [9], with $M_{12}(i,j)(\theta_\ell)=M_{12}(i,j)(\theta_{\ell i},\cdots,\theta_{\ell j-1})$.

There are two frequently used representations for C: the Christoffel's symbols (C will be denoted by C_C) or the M_D–notation [10] (C_D). In the first case

$$C_{kj}=\sum_{i=1}^{n}\frac{1}{2}\left[\frac{\partial M_{kj}}{\partial\theta_i}+\frac{\partial M_{ki}}{\partial\theta_j}-\frac{\partial M_{ij}}{\partial\theta_k}\right]\dot\theta_i. \tag{2.3}$$

For the M_D notation, first define a matrix M_D that depends on two vector arguments, with e_i the ith unit vector in $\mathbf{R}^n$:

$$M_D(\theta,v)\triangleq\sum_{i=1}^{n}\frac{\partial M(\theta)}{\partial\theta_i}ve_i^T \tag{2.4}$$

M_D satisfies the following properties:

$$M_D(\theta,v)w=\dot M(\theta,w)v,\ M_D^T(\theta,v)w=M_D^T(\theta,w)v. \tag{2.5}$$

$C_D(\theta,\dot\theta)$ can then be expressed as

$$C_D(\theta,\dot\theta)\triangleq M_D(\theta,\dot\theta)-\frac{1}{2}M_D^T(\theta,\dot\theta). \tag{2.6}$$

For the dynamical model (2.1), we have

$$C_D(\theta,\dot\theta)=\begin{bmatrix}C_{11}^A(\theta_\ell,\dot\theta_\ell)+C_{11}^B(\theta_\ell,\dot\theta_m) & C_{12}^A(\theta_\ell,\dot\theta_\ell)\\ C_{21}^A(\theta_\ell,\dot\theta_\ell) & 0\end{bmatrix}$$

$$\begin{aligned}C_{11}^A(\theta_\ell,\dot\theta_\ell) &= M_{D,11}^A(\theta_\ell,\dot\theta_\ell)-\frac{1}{2}{M_{D,11}^A}^T(\theta_\ell,\dot\theta_\ell)\\ C_{11}^B(\theta_\ell,\dot\theta_m) &= M_{D,11}^B(\theta_\ell,\dot\theta_m)-\frac{1}{2}{M_{D,11}^B}^T(\theta_\ell,\dot\theta_m)\\ C_{12}^A(\theta_\ell,\dot\theta_\ell) &= -\frac{1}{2}{M_{D,21}^A}^T(\theta_\ell,\dot\theta_\ell)\\ C_{21}^A(\theta_\ell,\dot\theta_\ell) &= M_{D,21}^A(\theta_\ell,\dot\theta_\ell)\end{aligned}$$

with obvious meaning of the expressions.

Note: C_{11}^A is the Coriolis and centrifugal term for the rigid robot; C_{11}^B is skew–symmetric with row m equal to zero and row i independent of any $\dot\theta_{m,j}$, $j\le i$, $i=1,\cdots,m-1$, $M_{D,21}^A$ is strictly lower triangular such that C_{12}^A is strictly upper triangular, and C_{21}^A is strictly lower triangular. The two representations (Christoffel's symbols and M_D–notation) for C yield a Coriolis and centrifugal matrix with the same structure.

Also $r(\theta)$ can be factored as

$$r(\theta)=\bar r(\theta)+R\theta \tag{2.7}$$

for any user defined constant matrix $R\in\mathbf{R}^{n\times n}$. We will partition R, r and $\bar r$ as

$$R=\begin{bmatrix}R_{11} & R_{12}\\ R_{21} & R_{22}\end{bmatrix},\ r(\theta)=\begin{bmatrix}r_1(\theta)\\ r_2(\theta)\end{bmatrix},\ \bar r(\theta)=\begin{bmatrix}\bar r_1(\theta)\\ \bar r_2(\theta)\end{bmatrix}$$

where R_{11}, R_{12}, R_{21}, $R_{22}\in\mathbf{R}^{m\times m}$ and r_1, r_2, $\bar r_1$, $\bar r_2\in\mathbf{R}^{m\times 1}$. For linear spring couplings a choice is

$$\bar r=\begin{bmatrix}g_\ell(\theta_\ell)\\ 0\end{bmatrix}\ R=\begin{bmatrix}K_eN^2 & -K_eN\\ -K_eN & K_e\end{bmatrix}\ K_e=\mathrm{diag}\{k_i\}$$

where k_i is the spring constant at joint i. The friction term D can also be decomposed in a linear and a nonlinear term as

$$D(\dot\theta)=D_0\dot\theta+D_1(\dot\theta).$$

Define the position and velocity errors as $\Delta\theta=\theta-\theta_d$, $\Delta\dot\theta=\dot\theta-\dot\theta_d$. The following bounds will be used in the next section.

$$\alpha_M=\inf_\theta\sigma_{min}\{M(\theta)\},\quad \alpha_p=\sigma_{min}\left\{R+BK_pB^T\right\}$$

$$\alpha_v=\sigma_{min}\left\{D_0+BK_vB^T\right\},\quad \gamma_d=\sup_{t\ge0}\left\|\dot\theta_d(t)\right\|$$

$$\gamma_{dd}=\sup_{t\ge0}\left\|\ddot\theta_d(t)\right\|,\quad \gamma_D=\sup_\theta\sum_{i=1}^{n}\left\|\frac{\partial M(\theta)}{\partial\theta_i}\right\|$$

$$\gamma_{Dd}=\sup_\theta\sum_{i=1}^{n}\sum_{j=1}^{n}\left\|\frac{\partial^2M(\theta)}{\partial\theta_i\partial\theta_j}\right\|,\quad \gamma_e=\sup_{t\ge0}\|\Delta\theta(t)\|$$

$$\gamma_M=\sup_\theta\|M(\theta)\|,\ \gamma_R=\sup_\theta\|\nabla_\theta\bar r(\theta)\|,\ \gamma_{D_0}=\|D_0\|$$

$$\gamma_v=\left\|D_0+BK_vB^T\right\|,\quad \gamma_p=\left\|R+BK_pB^T\right\|$$

where $K_p,K_v\in\mathbf{R}^{n\times n}$ are controller gains. For C represented using the Christoffel's symbols there exists a unique and finite real constant γ_C that satisfies, for any θ and any finite w, $\|C_C(\theta,w)\|\le\gamma_C\|w\|$. The evaluation of this bound is not straightforward. There also exists some constant γ_{Cd} such that

$$\left\|C_C(\theta_d,\dot\theta)-C_C(\theta,\dot\theta)\right\|\le\gamma_{Cd}\|\Delta\theta\|\left\|\dot\theta\right\|. \tag{2.8}$$

This bound is a function of the gradient norm supremum of C_C but is difficult to find analytically.

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Assume that, given $y_d = h(\theta_{\ell_d})$, we can solve for θ_d. There are many possible feedforward compensation based on the inverse dynamics of the system (**FF1** below) and its approximations that can be used

FF1. $Bu_{ff} = M(\theta_d)\ddot{\theta}_d + C(\theta_d, \dot{\theta}_d)\dot{\theta}_d + D(\dot{\theta}_d) + r(\theta_d)$

FF2. $Bu_{ff} = M(\theta_d)\ddot{\theta}_d + C(\theta_d, \dot{\theta}_d)\dot{\theta}_d + D(\dot{\theta}_d) + \bar{r}(\theta) + R\theta_d$

FF3. $Bu_{ff} = M(\theta)\ddot{\theta}_d + C(\theta, \dot{\theta})\dot{\theta}_d + D(\dot{\theta}_d) + r(\theta_d)$

FF4. $Bu_{ff} = M(\theta)\ddot{\theta}_d + C(\theta, \dot{\theta})\dot{\theta}_d + D(\dot{\theta}_d) + \bar{r}(\theta) + R\theta_d$

FF5. $Bu_{ff} = M(\theta)\ddot{\theta}_d + C_D(\theta, \dot{\theta})\dot{\theta}_d - \frac{1}{2}M_D(\theta, \dot{\theta})\dot{\theta}_d + \frac{1}{2}M_D(\theta, \dot{\theta}_d)\dot{\theta} + D(\dot{\theta}_d) + r(\theta_d)$

FF6. $Bu_{ff} = M(\theta)\ddot{\theta}_d + C_D(\theta, \dot{\theta})\dot{\theta}_d - \frac{1}{2}M_D(\theta, \dot{\theta})\dot{\theta}_d + \frac{1}{2}M_D(\theta, \dot{\theta}_d)\dot{\theta} + D(\dot{\theta}_d) + \bar{r}(\theta) + R\theta_d$

FF7. $Bu_{ff} = r(\theta_d)$

FF8. $Bu_{ff} = \bar{r}(\theta) + R\theta_d$

Some solutions to these inverse dynamics problems can be found in [7, 4].

3 Stability analysis under feedforward approximation

We first prove local exponential stability for two feedforward compensations **FF4** and **FF6**. The other types of feedforward are seen as perturbations to these canonical ones and bounds are found on these variations so to maintain stability, i.e. a robustness analysis with respect to variations of the feedforward is performed ([10] for the rigid robots case).

3.1 Exponential stability of canonical feedforward

We recall that the origin is an exponentially stable equilibrium point if there exists a scalar function $V_1(\Delta\theta, \Delta\dot{\theta})$, with $V_1(0,0) = 0$, and constants $(\alpha, \beta, \gamma) > 0$ such that

$$V_1 \geq \alpha \|x\|^2 \tag{3.1}$$

$$\dot{V}_1 \leq -\gamma \|x\|^2 \tag{3.2}$$

$$\|\nabla_x V_1\| \leq \beta \|x\| \tag{3.3}$$

where $x^T = [\Delta\theta^T, \Delta\dot{\theta}^T]$. (3.3) with $V_1(0,0) = 0$ imply that for some constant $\kappa > 0$, $V_1 \leq \kappa \|x\|^2$.

Consider the linear damping case, i.e. $D_1(\dot{\theta}) = 0$.

Assumption 3.1 *K_p and K_v are chosen such that $[R + BK_pB^T] > 0$ and $[D_0 + BK_vB^T] > 0$.*

Proposition 3.1 *Consider a flexible joint manipulator with linear damping. Assume that feedforward* **FF4** *or* **FF6** *is used. Close a motor position error $C_p(y_p) = K_pB^T\Delta\theta$ and a motor velocity error $C_v(B^T\Delta\dot{\theta}) = K_vB^T\Delta\dot{\theta}$ feedback. Then, for $\dot{\theta}_d$ sufficiently small in norm, the zero error equilibrium point is locally (to $\Delta\theta = 0$) exponentially stable.*

Proof : Choose as Lyapunov function candidate, where the last term is used to cancel extra terms introduced by the cross term in $\Delta\theta$ and $\Delta\dot{\theta}$,

$$V_1 = \frac{1}{2}\Delta\theta^T \left[R + BK_pB^T\right] \Delta\theta + c\Delta\theta^T M(\theta)\Delta\dot{\theta} + \frac{1}{2}\Delta\dot{\theta}^T M(\theta)\Delta\dot{\theta} + \frac{c}{2}\Delta\theta^T \left[D_0 + BK_vB^T\right] \Delta\theta \tag{3.4}$$

Evaluate the time derivative of V_1 along the solution of the system given by its closed-loop error equation,

$$\begin{aligned} \textbf{FF4}: \dot{V}_1 = & -c\Delta\theta^T \left[R + BK_pB^T\right] \Delta\theta \\ & + c\Delta\theta^T \left[\dot{M}(\theta, \dot{\theta}) - C_C(\theta, \dot{\theta})\right] \Delta\dot{\theta} \\ & - \Delta\dot{\theta}^T \left[D_0 + BK_vB^T - cM(\theta)\right] \Delta\dot{\theta} \end{aligned} \tag{3.5}$$

$$\begin{aligned} \textbf{FF6}: \dot{V}_1 = & -c\Delta\theta^T \left[R + BK_pB^T\right] \Delta\theta \\ & + c\Delta\theta^T [\dot{M}(\theta, \dot{\theta})\Delta\dot{\theta} - C_D(\theta, \dot{\theta})\Delta\dot{\theta} \\ & - \frac{1}{2}M_D(\theta, \dot{\theta})\dot{\theta}_d + \frac{1}{2}M_D(\theta, \dot{\theta}_d)\dot{\theta}] \\ & - \Delta\dot{\theta}^T \left[D_0 + BK_vB^T - cM(\theta)\right] \Delta\dot{\theta} \end{aligned} \tag{3.6}$$

Lower bound of V_1: [7].

$$\begin{aligned} V_1 \geq & \frac{1}{2}(\alpha_p + c\alpha_v) \|\Delta\theta\|^2 \\ & - c\gamma_M \|\Delta\theta\| \left\|\Delta\dot{\theta}\right\| + \frac{1}{2}\alpha_M \left\|\Delta\dot{\theta}\right\|^2 \\ \geq & \alpha_1 \|\Delta\theta\|^2 + \alpha_2 \left\|\Delta\dot{\theta}\right\|^2 \geq \alpha \|x\|^2 \end{aligned}$$

where we have written

$$\begin{aligned} -c\gamma_M \|\Delta\theta\| \left\|\Delta\dot{\theta}\right\| = & \frac{c\gamma_M}{2}[\left(\epsilon_1 \|\Delta\theta\| - \epsilon_1^{-1} \left\|\Delta\dot{\theta}\right\|\right)^2 \\ & - \epsilon_1^2 \|\Delta\theta\|^2 - \epsilon_1^{-2} \left\|\Delta\dot{\theta}\right\|^2] \end{aligned} \tag{3.7}$$

$\alpha_1 = \frac{1}{2}(\alpha_p + c\alpha_v - c\gamma_M\epsilon_1^2) > 0$, $\alpha_2 = \frac{1}{2}(\alpha_M - c\gamma_M\epsilon_1^{-2}) > 0$, and $\alpha = min\{\alpha_1, \alpha_2\}$. This is satisfied for c small enough, i.e. $0 \leq c < c_v$ where

$$c_v = \frac{\alpha_M\alpha_v}{2\gamma_M^2} + \left(\frac{\alpha_M^2\alpha_v^2}{4\gamma_M^4} + \frac{\alpha_M\alpha_p}{\gamma_M^2}\right)^{\frac{1}{2}}. \tag{3.8}$$

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Upper bound of V_1 : Similarly

$$\begin{aligned} V_1 \leq & \frac{1}{2}(\gamma_p + c\gamma_v)\|\Delta\theta\|^2 \\ & + c\gamma_M \|\Delta\theta\| \left\|\Delta\dot{\theta}\right\| + \frac{1}{2}\gamma_M \left\|\Delta\dot{\theta}\right\|^2 \\ \leq & \kappa_1 \|\Delta\theta\|^2 + \kappa_2 \left\|\Delta\dot{\theta}\right\|^2 \leq \kappa \|x\|^2 \end{aligned}$$

$\kappa_1 = \frac{1}{2}(\gamma_p + c\gamma_v + c\gamma_M \epsilon_2^2) > 0$, $\kappa_2 = \frac{1}{2}(\gamma_M + c\gamma_M \epsilon_2^{-2}) > 0$, $\kappa = max\{\kappa_1, \kappa_2\}$. Hence, V_1 is upper bounded for any bounded c.

Upper bound of $\nabla_{\Delta\dot{\theta}} V_1$: V_1 being lower and upper bounded by quadratic functions, and continuous, the norm of the gradient of V_1 with respect to its argument is also bounded. Evaluate explicitly the gradient of V_1 with respect to $\Delta\dot{\theta}$ from (3.4) :

$$\begin{aligned} \nabla_{\Delta\dot{\theta}} V_1(\Delta\theta, \Delta\dot{\theta}) &= cM(\theta)\Delta\theta + M(\theta)\Delta\dot{\theta} \\ \left\|\nabla_{\Delta\dot{\theta}} V_1\right\| &\leq \beta_1 \|\Delta\theta\| + \beta_2 \left\|\Delta\dot{\theta}\right\| \leq \beta \|x\| \end{aligned}$$

where $\beta_1 = c\gamma_M > 0$, $\beta_2 = \gamma_M > 0$, $\beta = \beta_1 + \beta_2$.

Upper bound of $\dot{V}_1$: for both **FF4** and **FF6**

$$\dot{V}_1 \leq -c\alpha_p \|\Delta\theta\|^2 + c\gamma_I \|\Delta\theta\| \left\|\Delta\dot{\theta}\right\|^2$$
$$+ c\gamma_I \|\Delta\theta\| \left\|\Delta\dot{\theta}\right\| \left\|\dot{\theta}_d\right\| - [\alpha_v - c\gamma_M] \left\|\Delta\dot{\theta}\right\|^2 \quad (3.9)$$

with $\gamma_I = \gamma_{C_2}$ for the Christoffel's representation and $\gamma_I = \frac{3}{2}\gamma_D$ for the M_D notation of C. Using an argument similar to (3.7), assuming that $0 < c < c_2$, $\|\Delta\theta\|$ is uniformly bounded above by γ_e and $\left\|\dot{\theta}_d\right\|$ by γ_d, with $c_2 = \frac{\alpha_v}{\gamma_M}$, one has

$$\dot{V}_1 \leq -\gamma_1 \|\Delta\theta\|^2 - \gamma_2 \left\|\Delta\dot{\theta}\right\|^2 \leq -\gamma \|x\|^2$$

where $\gamma_1 = c\left[\alpha_p - \frac{\gamma_I}{2}\epsilon_3^2 \gamma_d\right]$, $\gamma_2 = \alpha_v - c\gamma_M - c\gamma_I\gamma_e - c\frac{\gamma_I}{2}\epsilon_3^{-2}\gamma_d$, $\gamma = min\{\gamma_1, \gamma_2\}$, for $(\gamma_1, \gamma_2) > 0$. V_1 is negative definite if

$$\gamma_d < \frac{2}{\gamma_I}\left[\frac{\alpha_p(\alpha_v - c\gamma_M - c\gamma_I\gamma_e)}{c}\right]^{\frac{1}{2}} \quad (3.10)$$

Summary : Conditions for local exponential stability are satisfied if $\Delta\theta$ and $\dot{\theta}_d$ are small enough in norm and satisfy relation (3.10) for $0 < c < \min\{c_v, c_2\}$. We may ascertain a rate of convergence larger or equal to $\sigma = \frac{\gamma}{\kappa}$. Note that a better evaluation of the convergence rate is given by $\sigma = \min\left\{\frac{\gamma_1}{\kappa_1}, \frac{\gamma_2}{\kappa_2}\right\}$. Also note that (3.10) indicates that an increased supremum of the norm of $\dot{\theta}_d$ reduces the allowable bound γ_e such that the region of convergence decreases.

3.2 Procedure for generic feedforward

Write the different forms of feedforward, u_{ff}, in terms of the canonical feedforward, u_{ff_c}, and of a variation term, e, as $Bu_{ff} = Bu_{ff_c} + e$. Consider a scalar function $V_2 = V_1$. The time derivative of V_2 along the solution is

$$\dot{V}_2 = \dot{V}_1 + \left[\nabla_{\Delta\dot{\theta}} V_1\right]^T M^{-1}(v)e \quad (3.11)$$

where $M(v)$ is the multiplier of $\Delta\ddot{\theta}$ in the error equation of the system with the canonical feedforward, e.g. $M(\theta)$ for feedforward **FF4**. Write

$$\begin{aligned} \dot{V}_2 &\leq \dot{V}_1 + \left\|\left[\nabla_{\Delta\dot{\theta}} V_1\right]^T M^{-1}(v)e\right\| \\ &\leq \dot{V}_1 + \frac{1}{\alpha_M}\left\|\nabla_{\Delta\dot{\theta}} V_1\right\| \|e\| \\ &\leq -\gamma\|x\|^2 + \zeta\|x\|\|e\| \quad (3.12) \end{aligned}$$

being $\left\|\nabla_{\Delta\dot{\theta}} V_1\right\| \leq \|\nabla_x V_1\| \leq \beta\|x\|$ and $\zeta = \frac{\beta}{\alpha_M}$. Then, if the norm of the variation of the feedforward satisfies one of the two following bounds, we obtain stability of the system under the new feedforward.

Case 1: $\|e\| \leq \chi\|x\|$ for some constant $\chi > 0$. Then, using (3.12), we obtain

$$\dot{V}_2 \leq -(\gamma - \zeta\chi)\|x\|^2$$

Hence, exponential stability with rate of convergence $\sigma_1 = \frac{\gamma - \zeta\chi}{\kappa}$ is obtained if the variation of the feedforward is small enough in the sense that $\chi < \frac{\gamma}{\zeta}$.

Case 2: $\|e\| \leq \chi\|x\| + \rho$ for some $(\chi, \rho) > 0$.

$$\dot{V}_2 \leq -(\gamma - \zeta\chi)\|x\|^2 + \zeta\rho\|x\| \quad (3.13)$$

Define $\sigma_2 = \frac{(\gamma - \zeta\chi - \zeta\rho\epsilon^{-2})}{\kappa}$, $\rho_2 = \frac{\zeta\rho\epsilon^2}{4}$ and with an argument similar to (3.7) we get $\dot{V}_2 \leq -\sigma_2 V_2 + \rho_2$ and

$$V_2(t) \leq e^{-\sigma_2(t-t_0)}\left[V_2(t_0, \Delta\theta(t_0), \Delta\dot{\theta}(t_0)) - \frac{\rho_2}{\sigma_2}\right] + \frac{\rho_2}{\sigma_2}$$

Assume that $\sigma_2 > 0$ for $\epsilon^2 > \epsilon_m^2$ with $\epsilon_m^2 = \frac{\zeta\rho}{\gamma - \zeta\chi}$. Define the supremum of V_2 over t as

$$\overline{V}_2 = \max\left\{V_2(t_0, \Delta\theta(t_0), \Delta\dot{\theta}(t_0)), \frac{\rho_2}{\sigma_2}\right\}$$

Since V_2 is positive definite, this implies that the state remains bounded, i.e. we obtain Lagrange stability. Also, convergence to the set $\left\{\Delta\theta(t), \Delta\theta(t) : V_2 \leq \frac{\rho_2}{\sigma_2}\right\}$ with rate of at least σ_2 is guaranteed. Note that choosing a large ϵ guarantees a fast convergence to a large set, while with a smaller $\epsilon > \epsilon_m > 0$ convergence is slower but to a smaller set, i.e. closer to zero.

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3.3 Stability for other feedforwards

We will consider the variation terms e, for the different forms of feedforwards, with respect to both **FF4** (C using Christoffel's symbols) and **FF6** (C using M_D–notation).

Note: In order to obtain a better bound on the variation of $\dot{V}_2$, explicitly evaluate (3.12) using (3.9) :

$$\begin{aligned}\dot{V}_2 &= \dot{V}_1 + \left(c\Delta\theta^T + \Delta\dot{\theta}^T\right) e \\ &\leq \dot{V}_1 + \left(c\|\Delta\theta\| + \left\|\Delta\dot{\theta}\right\|\right)\|e\| \\ &\leq \dot{V}_1 + \beta'\|x\|\,\|e\|\end{aligned}$$

where we may use $\beta' = (c+1)$. We then identify ζ with β' in (3.12).

3.3.1 FF4 as canonical feedforward

The variation terms e with respect to the canonical feedforward **FF4** are

FF1 $e = [M(\theta_d) - M(\theta)]\,\ddot{\theta}_d + \left[C_C(\theta_d,\dot{\theta}_d) - C_C(\theta,\dot{\theta})\right]\dot{\theta}_d + \bar{r}(\theta_d) - \bar{r}(\theta)$

FF2 $e = [M(\theta_d) - M(\theta)]\,\ddot{\theta}_d + \left[C_C(\theta_d,\dot{\theta}_d) - C_C(\theta,\dot{\theta})\right]\dot{\theta}_d$

FF3 $e = \bar{r}(\theta_d) - \bar{r}(\theta)$

FF7 $e = -M(\theta)\ddot{\theta}_d - C_C(\theta,\dot{\theta})\dot{\theta}_d - D_0\dot{\theta}_d + \bar{r}(\theta_d) - \bar{r}(\theta)$

FF8 $e = -M(\theta)\ddot{\theta}_d - C_C(\theta,\dot{\theta})\dot{\theta}_d - D_0\dot{\theta}_d$

We did not consider **FF5** and **FF6** because C is not represented by the same notation as **FF4**. The norm of the variation e can be bounded above by

FF1 $\|e\| \leq \left(\gamma_D\gamma_{dd} + \gamma_{Cd}\gamma_d^2 + \gamma_R\right)\|\Delta\theta\| + \gamma_C\gamma_d\left\|\Delta\dot{\theta}\right\|$

FF2 $\|e\| \leq \left(\gamma_D\gamma_{dd} + \gamma_{Cd}\gamma_d^2\right)\|\Delta\theta\| + \gamma_C\gamma_d\left\|\Delta\dot{\theta}\right\|$

FF3 $\|e\| \leq \gamma_R\|\Delta\theta\|$

FF7 $\|e\| \leq \gamma_R\|\Delta\theta\| + \gamma_C\gamma_d\left\|\Delta\dot{\theta}\right\| + \left(\gamma_M\gamma_{dd} + \gamma_C\gamma_d^2 + \gamma_{D_0}\gamma_d\right)$

FF8 $\|e\| \leq \gamma_C\gamma_d\left\|\Delta\dot{\theta}\right\| + \left(\gamma_M\gamma_{dd} + \gamma_C\gamma_d^2 + \gamma_{D_0}\gamma_d\right)$

The conclusions on the stability and performance for the different forms of feedforward are summarized in Table 1. Achievable type of stability, and qualitative conditions on design parameters for stability, for fast rate of convergence (f.r.c.) and for the convergence to a small set (c.s.s.) (Lagrange stability) are listed in the table. To guarantee stability, we require that σ_1 or σ_2 is positive depending on the form of the bound on the error signal and that a faster rate of convergence is obtained as these variables increase. For the Lagrange stability, convergence to a smaller set about the zero error state is guaranteed as ρ_2 decreases.

	Stab. cond.: small	Stab. type	Cond. f.r.c.: small	Cond. c.s.s.: small
FF1	$\gamma_d, \gamma_{dd}, \gamma_R$	l.ex.	$\gamma_d, \gamma_{dd}, \gamma_R$	—
FF2	γ_d, γ_{dd}	l.ex.	γ_d, γ_{dd}	—
FF3	γ_R	l.ex.	γ_R	—
FF4	—	l.ex.	—	—
FF7	γ_d, γ_R	Lag.	$\gamma_d, \gamma_{dd}, \gamma_R$	γ_d, γ_{dd}
FF8	γ_d	Lag.	γ_d, γ_{dd}	γ_d, γ_{dd}

Table 1: Stability and performance analysis, vs. **FF4**

3.3.2 FF6 as canonical feedforward

The variation terms e with respect to the canonical feedforward **FF6** are

FF1 $e = [M(\theta_d) - M(\theta)]\,\ddot{\theta}_d - \frac{1}{2}M_D(\theta,\dot{\theta}_d)\dot{\theta} + \left[C_D(\theta_d,\dot{\theta}_d) - C_D(\theta,\dot{\theta}) - \frac{1}{2}M_D(\theta,\dot{\theta})\right]\dot{\theta}_d + \bar{r}(\theta_d) - \bar{r}(\theta)$

FF2 $e = [M(\theta_d) - M(\theta)]\,\ddot{\theta}_d - \frac{1}{2}M_D(\theta,\dot{\theta}_d)\dot{\theta} + \left[C_D(\theta_d,\dot{\theta}_d) - C_D(\theta,\dot{\theta}) + \frac{1}{2}M_D(\theta,\dot{\theta})\right]\dot{\theta}_d$

FF3 $e = \frac{1}{2}M_D(\theta,\dot{\theta})\dot{\theta}_d - \frac{1}{2}M_D(\theta,\dot{\theta}_d)\dot{\theta} + \bar{r}(\theta_d) - \bar{r}(\theta)$

FF4 $e = \frac{1}{2}M_D(\theta,\dot{\theta})\dot{\theta}_d - \frac{1}{2}M_D(\theta,\dot{\theta}_d)\dot{\theta}$

FF5 $e = \bar{r}(\theta_d) - \bar{r}(\theta)$

FF7 $e = -M(\theta)\ddot{\theta}_d - \frac{1}{2}M_D(\theta,\dot{\theta}_d)\dot{\theta} - D_0\dot{\theta}_d - \left[C_D(\theta,\dot{\theta}) - \frac{1}{2}M_D(\theta,\dot{\theta})\right]\dot{\theta}_d + \bar{r}(\theta_d) - \bar{r}(\theta)$

FF8 $e = -M(\theta)\ddot{\theta}_d - \frac{1}{2}M_D(\theta,\dot{\theta}_d)\dot{\theta} - D_0\dot{\theta}_d - \left[C_D(\theta,\dot{\theta}) - \frac{1}{2}M_D(\theta,\dot{\theta})\right]\dot{\theta}_d$

The norm of the variation e can be bounded above by

FF1 $\|e\| \leq \left(\gamma_D\gamma_{dd} + \frac{3}{2}\gamma_{Dd}\gamma_d^2 + \gamma_R\right)\|\Delta\theta\| + \frac{3}{2}\gamma_D\gamma_d\left\|\Delta\dot{\theta}\right\|$

FF2 $\|e\| \leq \left(\gamma_D\gamma_{dd} + \frac{3}{2}\gamma_{Dd}\gamma_d^2\right)\|\Delta\theta\| + \frac{3}{2}\gamma_D\gamma_d\left\|\Delta\dot{\theta}\right\|$

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$$\mathbf{FF3}\ \|e\| \le \gamma_R \|\Delta\theta\| + \gamma_D\gamma_d \left\|\Delta\dot{\theta}\right\|$$

$$\mathbf{FF4}\ \|e\| \le \gamma_D\gamma_d \left\|\Delta\dot{\theta}\right\|$$

$$\mathbf{FF5}\ \|e\| \le \gamma_R \|\Delta\theta\|$$

$$\mathbf{FF7}\ \|e\| \le \gamma_R \|\Delta\theta\| + \frac{3}{2}\gamma_D\gamma_d \left\|\Delta\dot{\theta}\right\| + \left(\gamma_M\gamma_{dd} + \frac{3}{2}\gamma_D\gamma_d^2 + \gamma_{D_0}\gamma_d\right)$$

$$\mathbf{FF8}\ \|e\| \le \frac{3}{2}\gamma_D\gamma_d \left\|\Delta\dot{\theta}\right\| + \left(\gamma_M\gamma_{dd} + \frac{3}{2}\gamma_D\gamma_d^2 + \gamma_{D_0}\gamma_d\right)$$

The conclusions on the stability and performance for the different forms of feedforward are summarized in Table 2. Achievable type of stability, qualitative conditions on design parameters for stability, for fast rate of convergence and for convergence to a small set (for Lagrange stability) are listed in the table.

	Stab. cond.: small	Stab. type	Cond. f.r.c.: small	Cond. c.s.s.: small
FF1	$\gamma_d, \gamma_{dd}, \gamma_R$	l.ex.	$\gamma_d, \gamma_{dd}, \gamma_R$	—
FF2	γ_d, γ_{dd}	l.ex.	γ_d, γ_{dd}	—
FF3	γ_d, γ_R	l.ex.	γ_d, γ_R	—
FF4	γ_d	l.ex.	γ_d	—
FF5	γ_R	l.ex.	γ_R	—
FF6	—	l.ex.	—	—
FF7	γ_d, γ_R	Lag.	$\gamma_d, \gamma_{dd}, \gamma_R$	γ_d, γ_{dd}
FF8	γ_d	Lag.	γ_d, γ_{dd}	γ_d, γ_{dd}

Table 2: Stability and performance analysis, vs. **FF6**

3.3.3 Summary

We see from Tables 1 and 2 that the conditions for stability and fast rate of convergence with the different forms of feedforward involve the upper bound on three parameters that can be affected by design :

- Bounds (γ_d, γ_{dd}) are made small by using a slow desired output trajectory (θ_{ℓ_d} and its higher derivatives).
- The bound γ_R is made small by choosing R giving the best fit of the gradient of the spring characteristic in the sense that $\sup_\theta \{\|\nabla_\theta r(\theta) - R\|\}$ is minimized for R in the set of matrices meeting the design requirements. For a linear spring characteristic, this leads to $\gamma_R = 0$ if we may choose $[r(\theta) - R\theta] = 0$. Then, odd and even numbered feedforward forms become equivalent.

4 Conclusion

In the case of a controller composed by a model–dependent feedforward and a model–independent feedback, for flexible joint robots, we have presented a stability analysis for different types of feedforward. In particular it has been pointed out how, for feedforwards which differ in the amount of required model information and state measurements, from the knowledge of a Lyapunov function for a specific choice of the feedforward, it is possible to deduce the type of stability, local exponential or Lagrange, of the other forms of feedforward without explicitly searching for new Lyapunov functions. These results, together with the asymptotic (global) stabilization ones [2, 6, 5, 7], constitute an interesting step in the control of flexible joint robots.

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